

Artificial Intelligence

Course Learning Outcome

CLO1	Explain the fundamental Artificial Intelligence concepts, principles, and methods (C2, PLO1)	Exam
CLO2	Explain suitable AI techniques for a given problem using appropriate framework (C5, PLO2)	Assignment
CLO3	Demonstrate domain knowledge in terms of knowledge representation for a case study (A3, PL08)	Assignment

Assignment Overview and Instructions

This assignment is designed to develop your ability to investigate an Artificial Intelligence (AI) topic from a research perspective, critically analyze existing knowledge, formulate research questions or problems, and communicate your work in the form of an academic research paper.

You are required to select a topic related to Artificial Intelligence (AI) and prepare a research paper by following **ONE** of the following two approaches.

Option A: Systematic Literature Review (SLR)

Conduct a Systematic Literature Review (SLR) on your selected AI topic.

Your study should systematically identify, select, analyze, compare, and synthesize relevant published research to provide a comprehensive understanding of the selected topic. The paper should address appropriate research questions and discuss the current state of research, existing approaches or techniques, key findings, limitations, research gaps, challenges, and potential future research directions.

The review should follow an appropriate and clearly explained systematic review methodology, including the literature search and study-selection process.

Option B: Problem–Solution Research Paper

Identify a specific and clearly defined problem within your selected AI topic and propose an appropriate AI-based solution to address that problem.

Your work should demonstrate your understanding and application of relevant AI concepts and techniques. Where appropriate, you should design, develop, implement, test, and evaluate the proposed solution using suitable methods and performance measures.

The paper should clearly explain the problem statement, motivation, related work, proposed methodology or solution, implementation/experimental setup, evaluation, results, discussion, limitations, and potential future improvements.

General Requirements

Regardless of the selected approach, your work should demonstrate appropriate academic and research practices. You are expected to:

1. Select an appropriate and sufficiently focused topic within the field of Artificial Intelligence.
2. Clearly define the scope, motivation, objective(s), and/or research question(s) of the study.
3. Follow either the SLR approach or the Problem–Solution approach described above.
4. Conduct an appropriate review of relevant and credible academic literature.
5. Demonstrate critical thinking and analysis rather than merely summarizing or describing information.
6. Clearly explain and justify the methods, approaches, or techniques used in the study.
7. Present and critically discuss your findings or results.
8. Identify relevant limitations, challenges, research gaps, and/or future directions, where applicable.
9. Use appropriate figures, tables, diagrams, algorithms, equations, experimental results, or other forms of evidence where relevant.
10. Properly cite all sources and provide a complete reference list using the required referencing style.

Provided Learning and Writing Materials

To support you in completing this assignment, the following materials will be provided:

- A paper template that must be used for preparing the final paper.
- Several sample research papers that may be used as references to understand the expected organization, scope, depth, presentation, and academic writing style.
- Guideline documents that provide additional guidance on how to prepare and write different sections of a research paper.

Students should carefully study these resources before and during the preparation of their papers.

The provided sample papers are intended only as references and examples. Their content must not be copied or closely reproduced.

The final paper must follow the provided template. Students should replace the instructional or example content in the template with their own work while maintaining the required structure and formatting.

Responsible Use of Generative AI and Academic Integrity

The use of Generative Artificial Intelligence (Gen-AI) is permitted responsibly as a learning and supporting tool.

For example, Gen-AI tools may be used to help you:

- understand difficult AI or research concepts;
- explore ideas and possible research directions;
- clarify terminology;
- learn how particular research methods work; or
- improve your understanding of academic writing.

However, Gen-AI must not be used as a substitute for your own thinking, analysis, or academic writing. Students must not directly copy and paste Gen-AI-generated content into the submitted paper or present AI-generated work as their own original work.

Your final submission must demonstrate your own understanding, critical thinking, analysis, interpretation, and academic writing. You remain fully responsible for the accuracy, originality, validity, citations, references, and overall quality of everything included in your submission.

All submitted papers will be evaluated by the lecturer and will be processed through Turnitin and/or other academic-integrity checking mechanisms to examine text similarity and indicators of Gen-AI-generated content.

Where an unacceptably high similarity score or Gen-AI indicator is detected, the lecturer may further investigate the work and may require the student to explain, defend, or provide supporting evidence for the submitted work.

Where inappropriate use of Gen-AI, plagiarism, or other forms of academic misconduct are established, the lecturer reserves the right to apply an appropriate penalty, which may include mark reduction or, in serious cases, a failing mark for the assignment, subject to the University's applicable academic-integrity policies and procedures.

Important: Gen-AI may support your learning, but it must not replace your own thinking, research, analysis, or writing.

End-of-Semester Research Seminar

As part of this assessment, all completed research papers will be presented at the end of the semester in a small-scale research seminar.

The seminar is intended to provide students with an opportunity to communicate their research in an academic setting and gain experience presenting and defending their work.

In addition to the students enrolled in this class, other students and/or lecturers from the University may be invited to attend the seminar.

During the seminar, each student will be expected to:

- Deliver a clear and professional presentation of the research.
- Explain the motivation, research problem/questions, methodology, findings/results, and contributions of the work.
- Demonstrate a clear understanding of the submitted paper.

- Answer questions related to the research and justify the decisions made during the study.
- Participate professionally in academic discussion and respond appropriately to feedback and questions from the audience.

Students should therefore prepare their papers with the understanding that they may be required to present and defend the work before an audience beyond the immediate class.

The seminar aims to provide practical experience in research communication, knowledge sharing, academic discussion, and responding to questions and feedback from a wider academic audience.

Expected Outcome

By the end of this assignment, you should deliver TWO items:

- **Research Paper:** have produced a well-structured AI research paper that demonstrates your ability to independently investigate an AI-related topic, critically engage with academic literature, formulate meaningful research questions or a specific problem, apply an appropriate research methodology, and communicate your findings effectively.
- **Oral Presentation:** Failure to attend the final assignment presentation may result in a mark of **ZERO (0)** for the assignment component.

Whether you choose the SLR or Problem–Solution approach, the emphasis is not simply on producing a paper, but on demonstrating your understanding of the research process and your ability to think, analyze, investigate, and communicate as a researcher.

All figures and tables must be properly cited according to the journal format. References in your journal paper must be published within the last five (5) years. Please, do consider citing at least 5 related-papers to your topic from the given list of references in the supplementary materials.

Handing Out: 17 September 2026
Submission Due: 9 November 2026

Supplementary Materials

The description of each of the two option-papers' structures are attached in a separated file named "***Attachement-1 - Description of TWO Paper Structures.doc***".

The format of the paper Must follow the format of journal article of JATI, which is attached in a separated file named "***Attachment-2 - JATI Template.doc***"

If you chose Option A, your complete literature review must be written as a journal-style paper with a length of 20 (Minimum) - 25 (Maximum) pages (including references).

Submission Requirements in Moodle

- (i) Documentation (upload into moodle as DOC **and** PDF)
- (ii) Zip file video of your presentation (upload into moodle a single Video/zip file)